Systems III – Information Systems

Setting up your enivorment/Nastavitev okolja

Reminder for the e-classroom/Opmink za e-učilnico

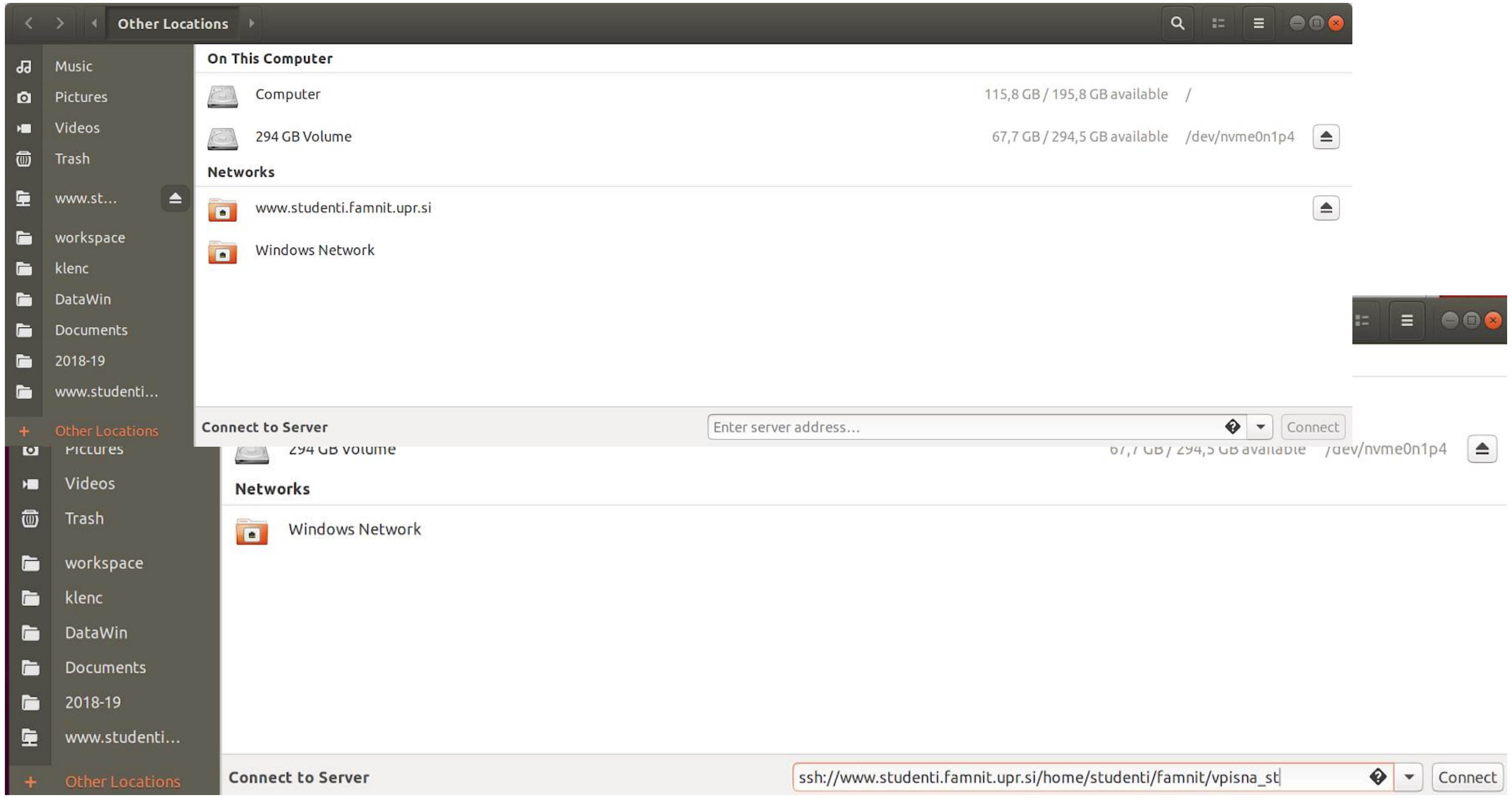
- We will be using only one e-classroom for all the groups, so make sure you are in.

- Uporabljali bomo samo eno e-učilnico za vse skupine, tako da prosim preverite, da ste vpisani v e-učilnico.

- https://e.famnit.upr.si/enrol/index.php?id=4941

Task 1- Linux platform

Step 1 Open files and click other locations

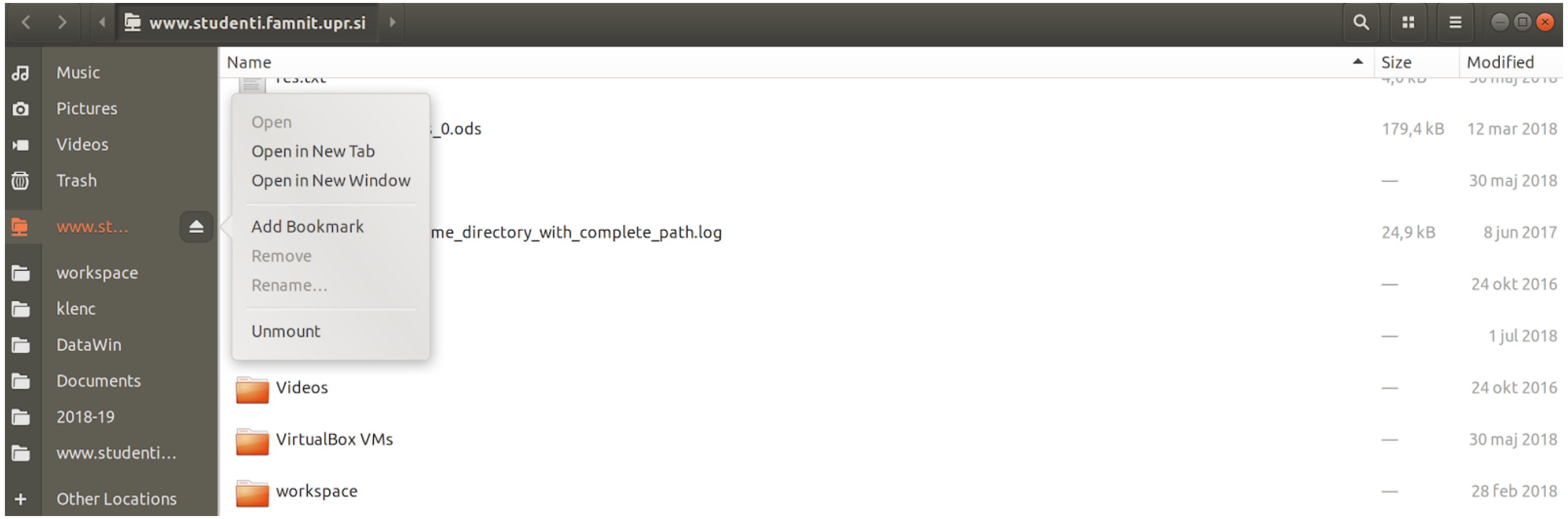


Step 2: In field connect to server enter following address: ssh://www.student.famnit.upr.si/home/studenti/famnit/ernolemnt_num

Task 1 - Linux Platform

- Step 3: Provide your enrolment number as username and use the standard password you use for logging into lab computers.
- Step 4: You should see your files now and can open them with the editor of your choice. You can add the connection to your bookmarks in order to speed up the connection process next time.

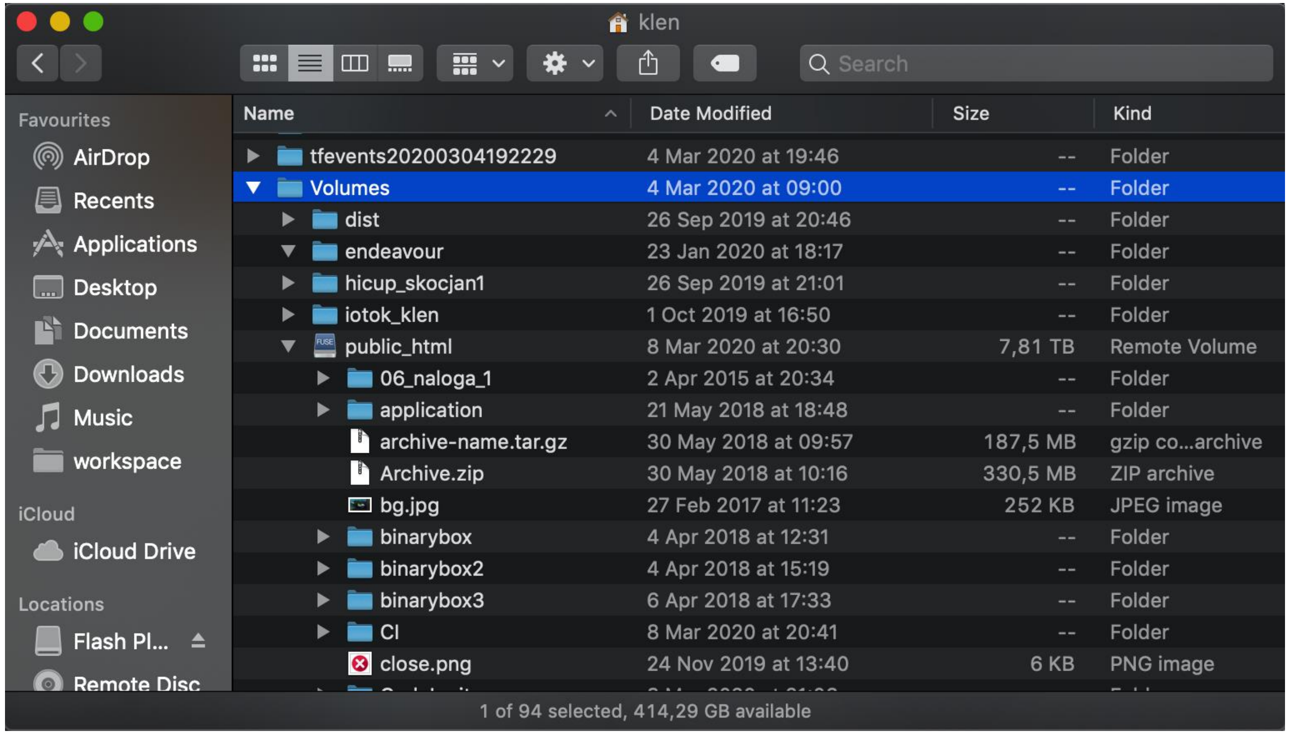
No need to install anything. Just open bash and run the following ssh command (remove <> should be removed)
$ ssh <enrolment_num>@www.studenti.famnit.upr.si



Task 1 - OS X – SSH into Uni

- Step1: Install sshfs https://howchoo.com/g/ymmxmzlmndb/how-to-install-sshfs
- Step2: mount as network volume using a command like
- $ 'mkdir /Users/guest/Volumes/server; sshfs klen@www.studenti.famnit.upr.si:/home/studenti/famnit/klen/public_html/ /Users/klen/Volumes/server -o volname=server,defer_permissions,follow_symlinks'
- Replace all red coloured text with your own data. Replace klen with enrolment_num.

Task 1 - OS X – SSH into Uni

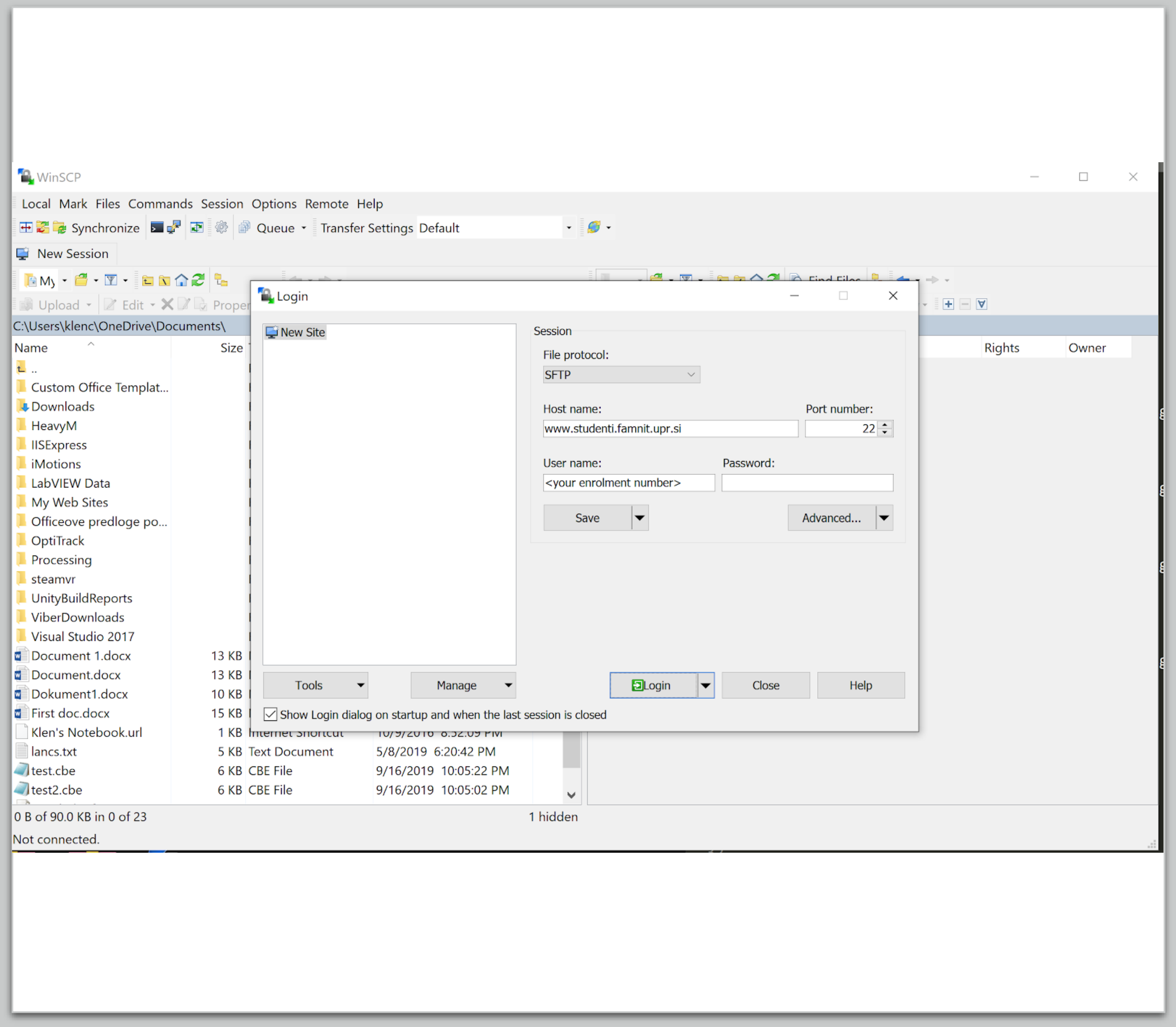


No need to install anything. Just open terminal and run the following ssh command (characters <> should be removed)

$ ssh <enrolment_num>@www.studenti.famnit.upr.si

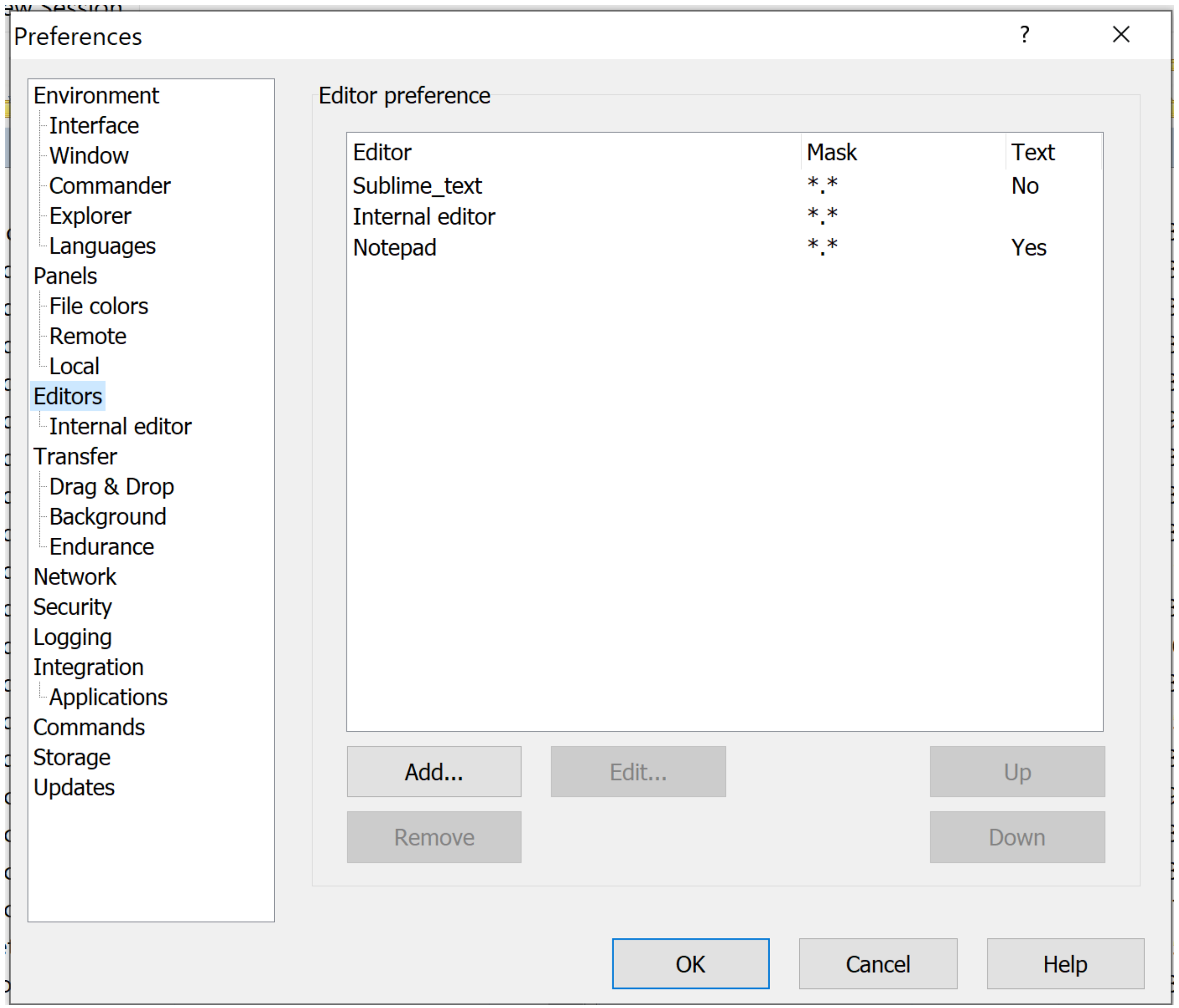
Task 1 - Windows – SSH into Uni

- **Step1: Install WinSCP from https://winscp.net/download/WinSCP-5.17.2-Setup.exe** or newer versions
- **Step2: Configure remote sftp connection:**



Windows platform

- **Step3: Configure your prefered text editor.**
- Go to Options/Preferences => Open Tab Editors and define the editor of your choice.
- Now you should be able to easily open and create files on your web server (e.g. public_html) directory.

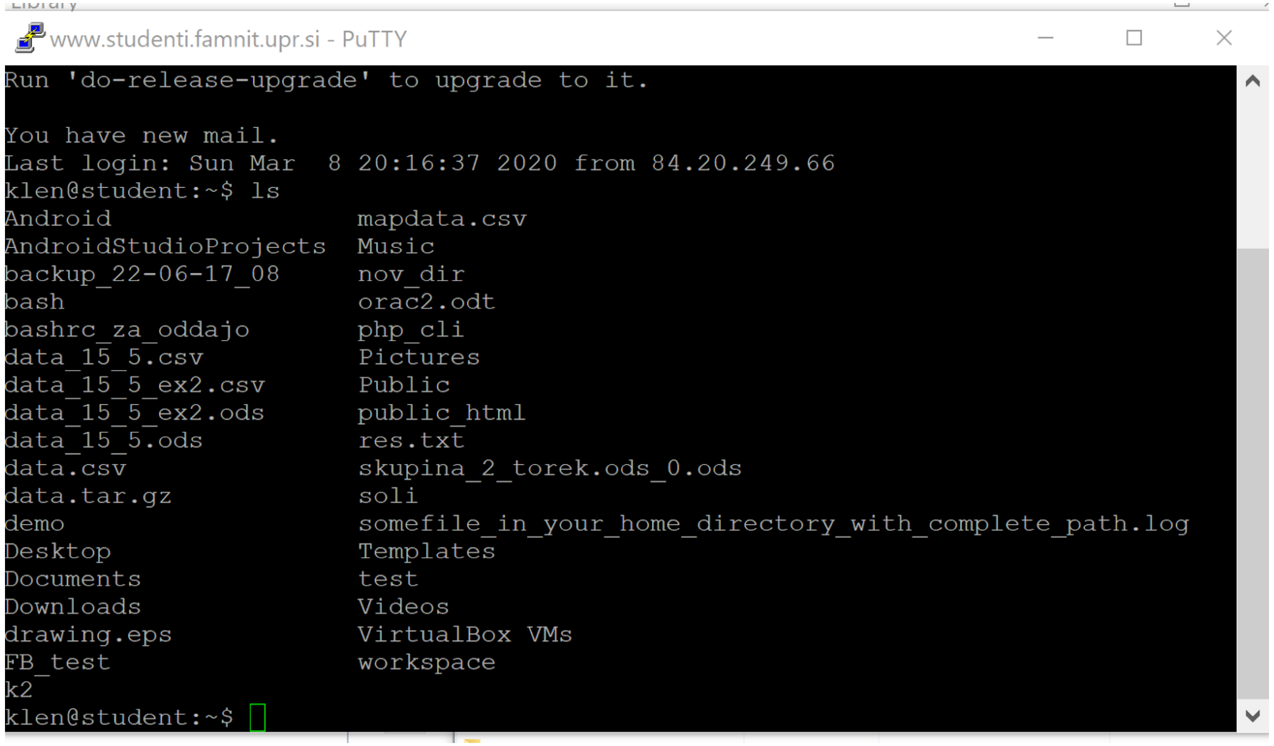


Windows platform – ssh client

- STEP1: Download an install Putty
- https://www.putty.org/

- STEP 2: Run putty and try to connect to www.studenti.famnit.upr.si

- This will open a shell command window prompting you to login. Provide your enrollment number and standard password you use for logging into lab computers.



ENGLISH INSTRUCTIONS

Task 2 – Creating an account at Pythonanywhere / Naloga 2 – Ustvari račun na Pythonanywhere

- Go to **https://eu.pythonanywhere.com/**
- And create an account and take a screenschot of the Main Dashboard with your account name visible.
- Upload the screenshot to the e-classroom

- Pojdi na **https://eu.pythonanywhere.com/**
- Ustvari račun in naredi screenschot Dashboarda z vidnim imenov vašega računa.
- Naloži nato na spletno učilnico

Task 3 – Creating a vitrual enviorment

- Go to the bash console in Pythonanywhere

$mkvirtualenv myvirtualenv - -python=/usr/bin/python3.10

- Once your virtualenv is ready and active, you'll see `(myvirtualenv) $` in your prompt.
- To check if its properly activated try the command $which pip
- And you should get a reponse like this:
- `/home/myusername/.virtualenvs/myvirtualenv/bin/pip`

- To reactivate your virtaul enviroment after the first use, you will need to use workon.
- `$ workon myvirtualenv`
- And deactivate to close it
- `$deactivate`

- Note Pythonanywhere uses a virtalenvwrapper to make it easier but there is small differentces in creating a basic virtual enviroment with virtualenv tool.

Task 4 - Installing Django – Part 1

- Now we are still in the console in our virtual environment and we will be downloading the required files to use the Django framework.
- $pip install django
- This should have installed Django 4.1.1
- Check if it worked using
- `$which django-admin`
- `$django-admin --version`
- Additonaly we will need to install a MySql packaged so that we will be able to use the UNIs existing MySQL database.
- $pip install mysqlclient
- Now create a project using
- $django-admin.py startproject <insert the name of your site>

Task 4 Instaling Django – Part 2

- Now go to the Web tab in Pythonanywhere.
- Select the "Add a new web app" option on the left-hand side.
- Select the "Manual configuration" option from the list.
- Click "Next" and wait for the system to tell you that the web app has been created.
- Now you can check that you have a web app -- go to *yourusername*.pythonanywhere.com in your browser to see the simple site thata was generated for you.
- Now open up the WSGI file for your web app and delete everything except the Django section and then uncomment the Django section.

Task 4

Your WSGI file should look something like this:

```
# +++++++++++ DJANGO +++++++++++
# To use your own django app use code like this:
import os
import sys

# assuming your django settings file is at '/home/myusername/mysite/mysite/settings.py'
path = '/home/myusername/mysite'
if path not in sys.path:
    sys.path.insert(0, path)

os.environ['DJANGO_SETTINGS_MODULE'] = 'mysite.settings'

# serve django via WSGI
from django.core.wsgi import get_wsgi_application
application = get_wsgi_application()

```

Task 4

- Then, back on the web tab itself, edit the path to your virtualenv in the Virtualenv section. You can specify the full path, */home/myusername/.virtualenvs/myproject*, or just the short name of the virtualenv, *myproject*, and the system will automatically expand it to the full path after you submit it.
- Add the full path where your project is under the source code section.
- Add "username.eu.pythonanywhere.com" in Allowed hosts in settings.py file that you will find in your project folder that was created by Django
- Save it, then go and hit the **Reload** button for your domain.
- Checking it worked.
- Go to site -- you should see the Django "it worked!" page.

Lab work that needs to be submited

- Task 1 - screenshot of you ssh into the server
- Task 2 – screenshot of the dashboard with username visible
- Task 3 – screenshot of a created and activated virtual environment
- Task 4 – screenshots of the installed packeges + changes in the WSGI files and the insert of your virtualenv file path in the web tab

- Task need to be submitted by the end of the week. This season is not graded, it's just so I know you are all properly setup.

SLOVENSKA NAVODILA

Naloga 3 – Ustvarjanje vitrualne okolje

- Pojdite na bash konzolo v Pythonanywhere
- $mkvirtualenv myvirtualenv - -python=/usr/bin/python3.10
- Ko je vaš virtualenv pripravljen in aktiven, boste videli (myvirtualenv) $ v vaši konzoli.
- Če želite preveriti, ali je pravilno aktiviran, poskusite z ukazom $which pip
- In moral bi dobiti tak odgovor.:
- `/home/myusername/.virtualenvs/myvirtualenv/bin/pip`

- Če želite po prvi uporabi ponovno aktivirati virtualno okolje uporabite workon.
- `$ workon myvirtualenv`
- In deactivate za zapiranje
- `$deactivate`

Naloga 4 - Inštalacija Djangta – del 1

- Zdaj smo še vedno v konzoli v našem virtualnem okolju in bomo prenesli potrebne datoteke za uporabo Django okvirja.
- $pip install django
- To bi moralo namestit Django 4.1.1
- Preverite, ali je delovalo z uporabo
- `$which django-admin`
- `$django-admin --version`
- Poleg tega bomo morali namestiti paket MySql, tako da bomo lahko uporabili obstoječo zbirko podatkov na Univerzi od MySQL.
- $pip install mysqlclient
- Zdaj ustvarite projekt z uporabo
- $django-admin.py startproject <vstavite ime po želji>

Naloga 4 - Inštalacija Djangta – del 2

- Zdaj pojdite na spletni zavihek v Pythonanywhere.
- Na levi strani izberite možnost »Dodaj novo spletno aplikacijo«.
- Na seznamu izberite možnost »Ročna konfiguracija«.
- Kliknite »Naprej« in počakajte, da vam sistem pove, da je spletna aplikacija ustvarjena.
- Zdaj lahko preverite, ali imate spletno aplikacijo -- pojdite na yourusername.pythonanywhere.com v brskalniku in si oglejte preprosto spletno mesto, ki je bilo ustvarjeno za vas.
- Zdaj odprite datoteko WSGI za vašo spletno aplikacijo in izbrišite vse razen razdelka Django in nato odkomentirajte razdelek Django.

Naloga 4 - Inštalacija Djangta – del 2

Datoteka WSGI bi morala izgledati takole:

```
# +++++++++++ DJANGO +++++++++++
# To use your own django app use code like this:
import os
import sys

# assuming your django settings file is at '/home/myusername/mysite/mysite/settings.py'
path = '/home/myusername/mysite'
if path not in sys.path:
    sys.path.insert(0, path)

os.environ['DJANGO_SETTINGS_MODULE'] = 'mysite.settings'

# serve django via WSGI
from django.core.wsgi import get_wsgi_application
application = get_wsgi_application()

```

Naloga 4 - Inštalacija Djangta – del 3

- Nato znova na samem spletnem zavihku uredite pot do virtualnega okolja v razdelku Virtualenv. Določite lahko celotno pot, /home/myusername/.virtualenvs/myproject ali samo kratko ime virtualenv, myproject in sistem ga bo samodejno razširil na polno pot, potem ko jo boste oddali.
- Dodajte celotno pot, kjer je vaš projekt v razdelku source kode.
- Dodajte »username.eu.pythonanywhere.com« v Dovoljeni gostitelji v settings.py datoteki, ki jo boste našli v mapi projekta, ki jo je ustvaril Django
- Shranite ga, nato pa pojdite in pritisnite gumb Za ponovno nalaganje domene.
- Preverite kako deluje.
- Pojdite na stran vašo stran ko prej omenjanea -- morali bi videti Django “it worked!" stran.

Laboratorijsko delo, ki ga je treba predložiti

- Naloga 1 - posnetek zaslona, ki ga ssh v strežnik
- Naloga 2 – posnetek zaslona nadzorne plošče z vidnim uporabniškim imenom
- Naloga 3 – posnetek zaslona ustvarjenega in aktiviranega navideznega okolja
- Naloga 4 – posnetki zaslona nameščenih paketov + spremembe v datotekah WSGI in vstavljanje virtualenv poti datotek v spletni zavihek

Nalogo je treba predložiti do konca tedna. Ta sezona ni ocenjena, ampak samo zato, da vem, da ste vsi pravilno nastavljeni.